

SECTION 01 / CASE OBJECTIVE

The objective of this case study is to explore how Quality Engineering principles can be applied to a large scale global payment ecosystem where security, transaction reliability, processing performance and customer experience must work together.

The central challenge is to strengthen protection against cyber threats and fraudulent activity while reducing the possibility of unnecessary impact on genuine transactions and maintaining a seamless customer experience.

The proposed framework uses TRIZ as the central problem solving perspective to address the contradiction between stronger security and lower customer friction.

SECTION 02 / INTEGRATED QUALITY PROBLEM

A global payment ecosystem must handle multiple quality requirements at the same time. Transactions should be secure against evolving cyber threats and fraudulent behaviour. At the same time, genuine transactions should not be unnecessarily blocked, transaction processing should remain reliable and consistent, and customers should experience a smooth payment journey.

Strengthening one requirement can create an unintended negative effect on another requirement. The Quality Engineering challenge is therefore not to optimise security, speed or customer experience independently. It is to develop an integrated quality approach that balances these requirements within one adaptive system.

SECURITY + FRAUD PROTECTION + TRANSACTION
RELIABILITY
+ PROCESSING PERFORMANCE + CUSTOMER EXPERIENCE
↓
INTEGRATED PAYMENT QUALITY

SECTION 03 / TRIZ & CONTRADICTION

TRIZ is used as the central problem solving approach because the payment quality challenge contains a fundamental contradiction. Increasing security controls may improve protection but can also increase friction for genuine customers.

Reducing security friction may improve the customer experience but can potentially weaken protection if controls are not intelligently designed. The objective is therefore to resolve the contradiction rather than simply accepting a trade off between security and customer experience.

STRONGER SECURITY
+ (CONTRADICTION) +
LOWER CUSTOMER FRICTION
↓
TRIZ CONTRADICTION RESOLUTION
↓
ADAPTIVE QUALITY SOLUTION

SECTION 04 / IDEAL FINAL RESULT

The ideal final result is a payment quality system that provides strong protection against suspicious activity while allowing genuine transactions to move through the payment experience with minimal unnecessary friction. The system should not treat every transaction identically.

Instead, security intensity should adapt according to relevant risk signals while maintaining reliability and customer experience as quality requirements. The objective is not to reduce security. The objective is to make security more adaptive and context aware.

LOW RISK → SEAMLESS PROCESSING
SUSPICIOUS → ADDITIONAL VERIFICATION
HIGH RISK → STRONGER SECURITY CONTROLS

SECTION 05 / ADAPTIVE RISK BASED QUALITY

The proposed solution combines the ideas of adaptive security, additional verification and real time response. Low risk transactions can follow a more seamless processing path.

When suspicious signals appear, an additional verification layer can be introduced rather than immediately blocking the transaction. When stronger risk indicators are present, the system can increase the intensity of security controls to protect the ecosystem while reducing unnecessary disruption.

REAL TIME SIGNALS → RISK ASSESSMENT
↓
ADAPTIVE SECURITY INTENSITY → TRANSACTION
DECISION

SECTION 06 / REAL TIME ADAPTATION

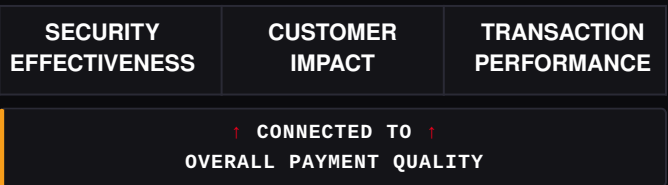
The proposed system continuously evaluates relevant transaction signals and adjusts the security response according to the observed risk level. The important principle is that security should not be static when the risk environment is dynamic.

A low risk transaction should not automatically experience the same level of intervention as a transaction showing multiple suspicious indicators. Adaptive response allows security controls to become more proportional to observed risk while maintaining the overall quality of the payment experience.

SECTION 07 / QUALITY MEASUREMENT

The success of the proposed framework should be evaluated through three connected quality dimensions. The first is security effectiveness, including the ability to detect and prevent suspicious or fraudulent activity. The second is customer impact, including whether genuine transactions are unnecessarily declined or disrupted.

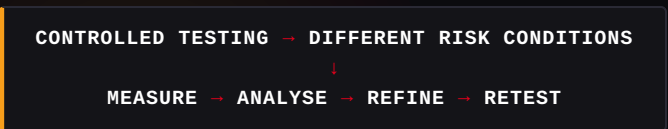
The third is transaction performance, including reliability and consistency of payment processing. These dimensions should be evaluated together because improving one dimension while damaging another would not represent overall quality improvement.



SECTION 08 / CONTROLLED VALIDATION

The proposed adaptive system should be validated under controlled conditions before broader implementation. Different transaction scenarios and relevant risk conditions should be evaluated to understand whether the adaptive response behaves as intended.

The validation should examine whether low risk transactions remain smooth, whether suspicious transactions receive appropriate additional verification and whether higher risk situations trigger stronger controls before wider implementation.



SECTION 09 / QUALITY & CX BALANCE

Security and customer experience should not be treated as completely independent objectives. The proposed framework attempts to achieve both through adaptive decision making. The system should focus on preventing unnecessary friction rather than reducing necessary security.

This creates a Quality Engineering perspective where the objective is to protect the customer and the payment ecosystem simultaneously. The quality goal is strong security combined with minimal unnecessary customer friction.

SECTION 10 / PDCA & CONTINUOUS IMPROVEMENT

After controlled validation, the improvement cycle continues through PDCA. Plan defines the quality problem, relevant risks, baseline and improvement objective. Do implements the adaptive approach under controlled conditions.

Check evaluates security effectiveness, genuine transaction impact and transaction performance. Act standardises validated improvements or restructures the approach when expected outcomes are not achieved. The process continues through Kaizen and continuous monitoring because cyber threats and transaction patterns change over time.



SECTION 11 / FRAMEWORK SUMMARY

Integrated Quality Problem → TRIZ Contradiction → Ideal Final Result → Adaptive Risk Based Response → Real Time Signal Analysis → Controlled Validation → PDCA & Kaizen.

SECTION 12 / KEY INSIGHT

A strong payment quality system should not force security and customer experience into a simple trade off. Quality is achieved when security effectiveness, transaction reliability, processing performance and customer experience are managed as interconnected quality requirements.

METHODOLOGY: TRIZ Contradiction Analysis, Ideal Final Result, Adaptive Risk Based Quality, Real Time Signal Analysis, Quality Measurement, Controlled Validation, PDCA, Kaizen, Continuous Improvement.

